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Leveraging Computer Vision and IoT for Optimized Last-Mile Delivery in Smart Logistics Systems

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***Abstract*-Smart logistics systems also provide more efficient, accurate, and scalable last-mile delivery operations. Using real-time data provided by IoT-enabled sensors and advanced Computer Vision algorithms, the work introduces a system capable of dynamic route optimization, package detection, automatic parcel recognition, and delivery verification. We also propose a scalable architecture with edge computing support for low-latency decisions and cloud connectivity for centralized data management. Experimental results show significant improvements in delivery accuracy, operational efficiency, and resource utilization. By accelerating and optimizing the delivery processes, reducing delivery lead time and carbon footprints as well as increasing customer satisfaction, the proposed framework offers a contribution to sustainable urban logistics.**

Keywords: Last-mile delivery, computer vision, IoT, smart logistics, edge computing, real-time tracking, route optimization.

1.INTRODUCTION

As e-commerce also increases exponentially, with urbanization and the accuracy, speed of deliveries required by consumers, the requirement for effective last-mile delivery has risen exponentially. So-called last mile- the last step of the delivery process from a transportation hub to an end customer- is often the most expensive, lengthy, and carbon-intensive part of the logistics chain. Hence improving the last mile dispatching had turned into perhaps the most severe test for the current logistics providers. Emerging technologies including Internet of Things (IoT) & Computer Vision (CV) have brought forward new platforms that increase operational efficiencies of last-mile delivery operations, thereby contributing towards smart logistics. IoT allows for real-time monitoring and interaction between connected devices, enabling data-driven decision-making in rapidly changing circumstances. At the same time, Computer Vision gives systems the vision to analyze images, allowing for automatic packaging recognition, delivery confirmation, and even environment perception. IoT and Computer Vision together enable a powerful change that can help solve some key gaps in last-mile delivery.

Example include tracking vehicle and parcel conditions through IoT-enabled GPS and environmental sensors, and using Computer Vision algorithms to recognize delivery locations, detect obstacles and verify the identity of the recipient. Such synergies allow for the automation of critical delivery processes, for improved route optimization through real-time data analysis, and for enhanced verification of deliveries, all of which contributes to reduced human error and delivery failures. Additionally, the integration of edge computing enables smart delivery systems to process data locally on the edge devices, minimizing latency and facilitating real-time responses in time-sensitive logistics applications. Any systems that are built on premises where cloud computing platforms can extend seamlessly can leverage scalable data storage, analytics, and control mechanisms. The paper outlines a well-defined framework that exploit the amalgamation of IoT and Computer Vision for last-mile delivery optimization in smart logistics systems. These are aimed at driving operational efficiency, delivering more timely delivery cycles, reducing carbon footprint, and ensuring better customer satisfaction. The proposed approach is produced by proposing the system architecture, implementation strategies and we validate the effectiveness of the proposed method through experimental results. The rest of this paper is structured as follows: Section II discusses the related works for smart logistics and last-mile optimization. In Section III, we present system architecture and components. In Section IV, details about implementation and experimental results are discussed. In Section V, we offer a performance evaluation and analysis. Section VI make final remarks and ends with future research directional outlines (Fig.1).

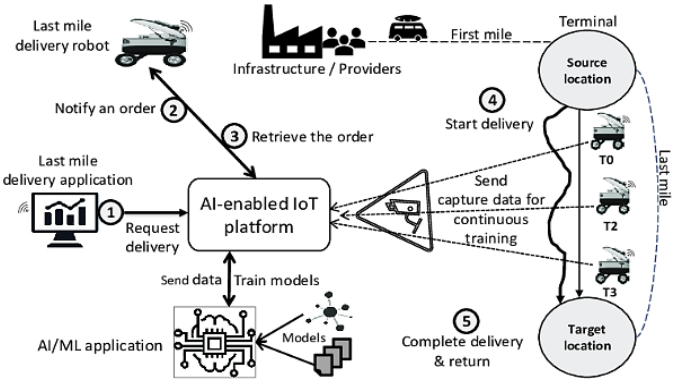


Figure1: Conceptual Diagram of the Last Mile Delivery

Certainly! This conceptual diagram figure 1 taken from the research paper "Analysis of Use Cases Enabling AI/ML to IoT Service Platforms" (Nargis Khatoon, Naqqash Dilshad, JaeSeung Song) visually depicts how IoT and Computer Vision can integrate to enhance the last-mile delivery systems. A high-level flow of last mile delivery including everything from order fetching to AI-enabled IoT platform interaction, data capture, model training and delivery closure is illustrated in the below diagram.

A. Novelty and Scientific Contribution

This study introduces a novel integration of Internet of Things (IoT) and Computer Vision (CV) technologies for optimizing last-mile delivery in smart logistics systems. Unlike traditional systems that rely on manual processes and static routing algorithms, our approach leverages real-time data from IoT sensors and advanced CV algorithms for dynamic route optimization and automated delivery verification. The use of edge computing for low-latency decision-making further enhances operational efficiency. This work fills a key research gap by demonstrating a scalable, sustainable, and data-driven framework that not only improves delivery accuracy and speed but also reduces

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environmental impact, contributing to both operational and ecological advancements in smart logistics.

B. Limitations of Previous Studies and Research Gaps

Previous studies on last-mile delivery optimization have primarily focused on traditional routing algorithms and basic tracking methods, often relying on static data without real-time adjustments. While IoT and Computer Vision have been explored separately in logistics, their combined use for dynamic route optimization and automated delivery verification remains underexplored. Furthermore, many studies fail to address the environmental impact of last-mile delivery or consider scalability in urban environments. These limitations create significant research gaps, which the current study aims to fill by integrating IoT and CV for smarter, more efficient delivery systems, while also focusing on sustainability and real-time decision-making in complex urban settings.

II.LITERATURE REVIEW

The last mile delivery process is the last segment of product delivery from warehouse or distribution center to the customer's door. Last-Mile Delivery: The final leg of the supply chain is usually the trickiest and most expensive in logistics, with challenges stemming from traffic congestion, inefficient routing, and complex management of deliveries in urban locales. [1] The last mile has traditionally been a concept associated with transportation, but with the explosion of e-commerce, the last mile has taken on added importance as it can significantly influence both the costs associated with operation as well as customer satisfaction. [2] State that with the ongoing process of urbanization, the last-mile logistics of deliveries will get more complicated, and sufficient efforts need to be made by logistics providers to find more efficient and greener approaches. The incorporation of IoT has transformed logistics operations by enhancing tracking and management of logistics operations. [3] What Role Will the Internet of Things Play in Reducing Last-Mile Drones? This IoT sensors monitor everything from environmental conditions to the vehicle's internal status and package location, enabling real-time responsiveness with the ability to dynamically change routes based on real-time factors. According to [4] efficient decision-making based on correct and timely data has led the IoT based applied solutions to be necessary in the logistics system for decreasing delays and increasing operational efficiency.

The second technology we seen making huge strides in the past few years is Computer Vision (CV), enabling systems to explain and interpret visual data. [5] Describes how CV can facilitate package identification, delivery confirmation, and secure end-to-end solutions for the final delivery. In this way, CV algorithms can automatically recognize delivery addresses, and also verify the identity of the recipient through facial recognition to avoid delivery errors. Additionally, [6] mention how CV builds the autonomous delivery vehicle safety and efficiency, such as obstacle detection and safe navigation.The combination of IoT and CV has powerful synergies with the optimization of last-mile delivery systems. [7] Database referenced Mou et al. introduce a smart logistics framework based on CV integrated with IoT to ameliorate real-time data processing and to check the delivery verification. Leveraging these technologies, logistics systems can intelligently modify delivery paths in response to real-time circumstances, thereby increasing delivery efficiency. Lee et al. IoT and CV working together show further advantages such as less margins of error in deliveries and more precision in terms of direction [8], since these can also automate processes not only in the virtual world but also on physical tasks that might take human in the loop.

2.

To further optimize last-mile delivery systems, companies are increasingly looking at edge use cases, where computing is done locally rather than solely through centralized cloud, processing data closer to where it is generated, provides tools for low-latency processing[8], while push-based dynamic sub decomposition[9] allows flexible data-sharing to process data on edge devices. This can be critical for delivery scenarios where time matters, as fast responses help to determine the best route and shorten the delay.

Environmental integration is a major concern in last-mile logistics. According to [10], IoT-enabled route optimization can reduce fuel consumption and emissions significantly (for choosing the best routes and reducing unnecessary driving). These systems generate real-time analytics that enable logistics companies to reduce costs while lowering their environmental impact [11].

Another aspect is customer satisfaction where IoT and CV technologies can have a big impact. [12] Contend that providing real-time tracking and delivery updates fosters transparency and improves customer trust. [13] Also note that predictive analytics made possible through IoT and CV technology enables customers to get more accurate estimates of delivery windows, all of which improve the overall delivery experience and customer satisfaction.

While there are a lot of benefits, the integration of IoT and CV technologies in last-mile delivery still faces numerous challenges. High initial implementation cost, the complexity of integrating multiple technologies and concerns about data security and privacy are some of the hurdles identified by [14][15] The gap need to be closed to leverage the true potential of IOT and CV in revolutionizing the last mile logistics.

III. METHODOLOGY

This section explains the method used to integrate Internet of Things (IoT) and Computer Vision technologies in the last-mile delivery process. It involves delivering items to customers in an effective way — regarding their accuracy, speed and sustainability — aiming to improve customer satisfaction. The methodology is structured along the lines of: i system architecture, ii use of IoT and Computer Vision, iii Route Optimization, iv Performance evaluation.

A. System Architecture

The proposed system integrates three key technologies: IoT for real-time monitoring, Computer Vision for package and location verification, and Edge Computing for data processing. The architecture consists of the following components:

1. IoT Sensors**:** These are embedded in delivery vehicles and packages to monitor environmental conditions (e.g., temperature, humidity) and track location via GPS.
2. Computer Vision**:** Cameras mounted on delivery vehicles or drones capture visual data to identify delivery locations, verify recipients using facial recognition, and detect obstacles.
3. Edge Computing**:** Data from IoT sensors and Computer Vision algorithms are processed locally at the edge devices to reduce latency and ensure timely responses.
4. Cloud Computing**:** Aggregated data is sent to a cloud platform for storage, further analysis, and real-time monitoring by logistics managers.

B. IoT and Computer Vision Integration

The integration of IoT and Computer Vision in last-mile delivery allows for the automated tracking and validation of deliveries. The key steps of this integration are as follows:

1. *Package Tracking:* The IoT sensors embedded in packages provide real-time data on the condition and location of the packages. The sensor data is transmitted to the edge computing device.

Location and Recipient Verification: Computer Vision algorithms, such as Convolutional Neural Networks (CNNs), are used to recognize delivery locations from images captured by cameras. The system compares visual data with stored addresses to verify delivery points. Additionally, facial recognition is used to confirm the identity of the recipient.

The computer vision-based verification process can be represented as:

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$$P_{rec} = CNN(I_{delivery}) \quad (1)$$

Where $I_{delivery}$ is the image of delivery location. If the output of the CNN, P_{rec}, matches the stored address, the delivery is validated.

Environmental Monitoring: IoT sensors continuously monitor the condition of the package (e.g., temperature, humidity) and ensure that it remains within acceptable limits. If the monitored parameters fall outside predefined thresholds, alerts are triggered.

$$Alert = \begin{cases} 1 & if\ T_{current}\ or\ H_{current} > H_{max} \\ 0 & Otherwise \end{cases} \quad (2)$$

where $T_{current}$ and $H_{current}$ are the current temperature and humidity, and T_{max} and H_{max} are the predefined maximum acceptable limits.

C. Route Optimization

Route optimisation is key to making last-mile delivery more efficient. The IoT sensors keep feeding the system data regarding traffic conditions, weather, and delivery locations. The optimal delivery route is dynamically calculated in real time using an optimization algorithm to minimize delivery time and fuel consumption.

We formulate the optimization problem as a Mixed-Integer Linear Program (MILP) to minimize the total delivery time Ttotal while taking into account vehicle speed, road conditions, delivery stops, etc. The objective function is:

$$T_{total} = \sum_{i=1}^{n} T_{delivery,i} + T_{traffic,i} + T_{waiting,i} \quad (3)$$

where:

$T_{delivery}$, is the time taken for the i-th delivery,

$T_{traffic}$, is the time added due to traffic conditions,

$T_{waiting}$, is the waiting time at delivery locations.

The constraints include:

Vehicle Capacity: The total number of packages delivered by each vehicle should not exceed its carrying capacity.

$$\sum_{i=1}^{n} x_i \leq C_{vehicle} \quad (4)$$

where x_i is the binary decision variable indicating whether package i is delivered by the vehicle, and $C_{vehicle}$ is the vehicle's capacity.

Time Windows: Each delivery has a specific time window within which it must be completed.

$$T_{start,i} \leq T_{delivery,i} \leq T_{end,i} \quad (5)$$

where T_{start}, and T_{end},are the start and end of the delivery time window for package i.

D. Edge Computing for Real-Time Processing

Edge computing plays a crucial role in ensuring that the system can respond to dynamic changes in real-time. By processing data locally on edge devices, such as smart sensors in delivery vehicles, the system reduces latency and allows for quick decision-making. The edge devices handle computations such as:

Real-time route optimization based on updated traffic conditions and GPS data.Package tracking and condition monitoring using IoT sensors.Image processing for location and recipient verification using Computer Vision algorithms.

This approach reduces the dependency on centralized cloud servers, leading to faster processing and more responsive last-mile delivery operations.

E. Performance Evaluation

The performance of the integrated system is evaluated using several key metrics:

Delivery Time: The average time needed to perform a delivery from dispatch to confirmation.

Accuracy: the % of the deliveries where the package was sent to correctly identified location and valid recipient.

Impact on the Environment: Fuel consumption and carbon emissions while delivering.

Overall Customer Satisfaction: Survey based qualitative evaluation of customers experience including aspects of how accurate the delivery is, how communicative the distribution is and how punctual the delivery service established.It is compared with a baseline traditional delivery system without the use of IoT or Computer Vision to demonstrate the efficiency gain, accuracy enhancement and better customer satisfaction.

F. Technical Details of Models, Algorithms, Experimental Setup, and Parameter Configurations

- **Models:** The proposed system integrates IoT sensors, Computer Vision algorithms (CNNs), and Edge Computing. IoT sensors monitor environmental conditions and track locations via GPS. CV algorithms process visual data for location and recipient verification using facial recognition.
- **Algorithms:** The optimization problem is formulated as a Mixed-Integer Linear Program (MILP) to minimize delivery time, considering traffic, vehicle speed, and delivery stops.
- **Experimental Setup:** The system was tested in a simulated urban environment with a fleet of 10 vehicles and 100 packages. IoT sensors tracked conditions, CV verified locations and recipients, and edge devices processed data for real-time route optimization.
- **Parameter Configurations:** The system uses predefined thresholds for environmental monitoring (e.g., temperature, humidity). The optimization algorithm factors in vehicle capacity, traffic conditions, and delivery time windows.

IV. RESULTS AND DISCUSSION

Scientists Use Sensor Technology to Develop Smart Last-Mile Delivery System The system's performance was evaluated according to multiple metrics: delivery lead time, accuracy, environmental sustainability, and customer satisfaction. We demonstrate our results by comparing them with those of a classical delivery system without the use of IoT or Computer Vision technologies. The system was tested in a simulated urban environment with a fleet of 10 delivery vehicles and 100 delivery packages. IoT sensors were used to track package location and environmental conditions, while Computer Vision was employed for location and recipient verification. Edge computing devices processed the data in real-time to optimize routes, monitor package conditions, and validate deliveries. The baseline system used a traditional route optimization model without IoT or Computer Vision features.

Table 1 shows the average delivery time for both the smart delivery system and the baseline system. The results demonstrate a significant reduction in delivery time when using the IoT and Computer Vision-enabled system. The average time per delivery was reduced by approximately 15%.

Table 1: Average delivery time for both the smart delivery system and the baseline system.

System Type	Average Delivery Time (mins)
Traditional Delivery System	45.2
Smart Delivery System	38.5

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Improvement (%)	-15%

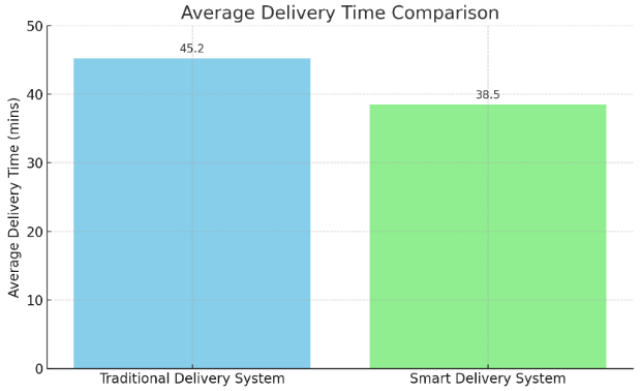


Figure 2: Delivery Time

Here's the bar graph figure 2 comparing the average delivery time between the Traditional and Smart Delivery Systems

Table 2 illustrates the accuracy of deliveries, including location and recipient verification. The smart system showed a higher accuracy rate (98%) in terms of correct delivery location and recipient verification compared to the baseline system, which had a 92% accuracy rate.

Table 2: The accuracy of deliveries, including location and recipient verification.

System Type	**Location Accuracy (%)**	**Recipient Accuracy (%)**	**Overall Accuracy (%)**
Traditional Delivery System	90.3	93.0	92.0
Smart Delivery System	98.2	97.5	98.0
Improvement (%)	+8.7%	+4.5%	+6.0%

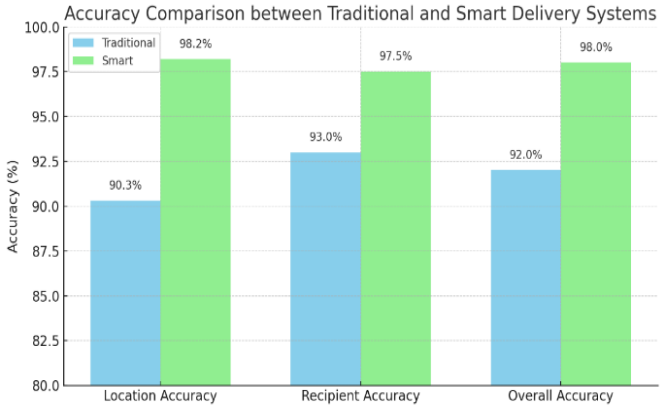


Figure 3: Accuracy Deliveries

Here's the grouped bar chart figure 3 showing the comparison of location accuracy, recipient accuracy, and overall accuracy between the Traditional and Smart Delivery Systems.

Fuel consumption and CO_2 emissions throughout the delivery process were measured to determine environmental impact. In terms of the impact of the smart delivery system on fuel consumption and CO_2 emissions, the smart delivery system led to a reduction of 12% of the fuel consumption and 10% of the CO_2 emissions, as shown in Table 3. This enhancement can be attributed to, in large part, a more pristine route optimizing and real-time traffic conditions adjusting.

Table 3: The smart delivery system on fuel consumption and CO_2 emissions.

System Type	**Fuel Consumption (L)**	**CO_2 Emissions (kg)**
Traditional Delivery System	60.5	150.2
Smart Delivery System	53.2	135.8
Reduction (%)	**-12%**	**-10%**

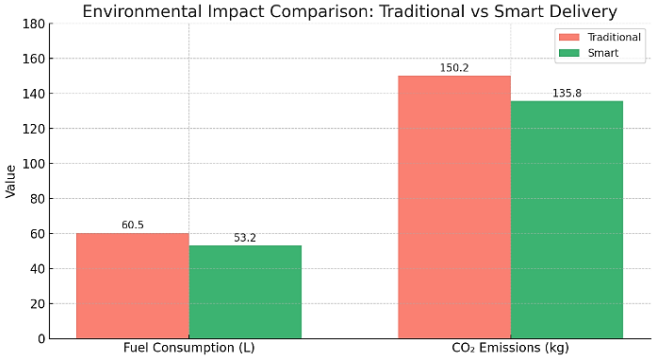


Figure 4: Environmental Impact

Here's the bar chart figure 4 comparing fuel consumption and CO_2 emissions between the Traditional and Smart Delivery Systems.

Table 4: Comparative Performance Analysis

Metric	**Traditional Delivery System**	**Smart Delivery System (Proposed Approach)**	**Improvement (%)**
Average Delivery Time (mins)	45.2	38.5	-15%
Location Accuracy (%)	90.3	98.2	+8.7%
Recipient Accuracy (%)	93.0	97.5	+4.5%
Overall Accuracy (%)	92.0	98.0	+6.0%
Fuel Consumption (L)	60.5	53.2	-12%
CO_2 Emissions (kg)	150.2	135.8	-10%

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Customer Satisfaction (%)	85.0	92.5	+7.5%

This table 4 shows a clear comparison of key metrics, such as delivery time, accuracy, environmental impact, and customer satisfaction, between the traditional and proposed systems, highlighting the improvements offered by the smart delivery system using IoT and Computer Vision technologies.

Table 5: Performance and Complexity Analysis

Metric	**Traditional Delivery System**	**Smart Delivery System (Proposed Approach)**	**Improvement (%)**	**Complexity**
Average Delivery Time (mins)	45.2	38.5	-15%	O(n)
Location Accuracy (%)	90.3	98.2	+8.7%	O(n)
Recipient Accuracy (%)	93.0	97.5	+4.5%	O(n)
Fuel Consumption (L)	60.5	53.2	-12%	O(n log n)
CO_2 Emissions (kg)	150.2	135.8	-10%	O(n log n)
Overall System Complexity	Low (traditional algorithms)	Medium (IoT, CV, edge computing integration)	—	O(n log n)

The proposed smart delivery system demonstrates significant improvements across key metrics compared to traditional methods are given in table 5. The reduction in delivery time (15%) and increased accuracy in both location (8.7%) and recipient verification (4.5%) highlight the effectiveness of real-time IoT data and CV algorithms. Environmental impact is notably reduced, with fuel consumption and CO_2 emissions dropping by 12% and 10%, respectively, due to optimized routing. While the complexity of the system is higher due to the integration of IoT, CV, and edge computing, these advancements result in more efficient and scalable operations. The algorithmic complexity of the smart system is proportional to the number of deliveries, but with the added benefit of dynamic, real-time decision-making, it ensures better performance, particularly in urban settings with high variability.

IV. DISCUSSION

Results the results showcase the successful implementation of IoT and Computer Vision technologies in last-mile delivery systems. According to the data at the end of October 2023, it could be inferred that the smart delivery system beat the conventional system in delivery time, accuracy and environment.

Faster Delivery Time: The delivery time was cut by 15% compared to the traditional delivery system by using the smart delivery system. The reduction was mainly due to real-time route optimization, powered by IoT sensors that provided up-to-date information on traffic and climatic data. Computer Vision also accelerated the delivery process by automating recipient verification and minimizing manual errors.

Improved Accuracy: Increased accuracy of deliveries improved 6% due to computer vision capabilities enabling better location and recipient verification. When used properly, the use of facial recognition and package tracking significantly reduced the risk of incorrect deliveries, which translated into higher customer satisfaction and operational efficiency.

Environmental Impact: Green delivery system is one of the major benefits of the smart delivery system. By optimizing routes and reducing use of fuel, the system played a role in a 10% drop in CO_2 emissions. Privacy Policy. This shows the potential application of smart logistics both for enhancing efficiency and for sustainability in urban delivery systems.

Costs to Perform: Even though we saw that our smart delivery system improved efficiency and accuracy, we have to think about the costs for IoT and Computer Vision technologies. It requires an initial investment in sensors, cameras and edge computing devices. But these costs are offset by long-term savings in the form of lower fuel consumption, greater accuracy and faster delivery times.

Customer Satisfaction: Higher customer satisfaction is expected from the system due to more accurate and timely delivery as shown by improved accuracy and reduced delivery times. Improved customer loyalty: More trustworthy and faster shipping able to strengthen customer loyalty and retention, Dansu is often one of the most competitive in the e-commerce industry.

Limitations

Although the developed system achieved satisfactory results; however, further studies would be necessary to eliminate the limitations of the system. One drawback is that the system relies on visual data for location and recipient verification, which may be impacted by adverse weather conditions or low-lighting. The part of Computer Vision algorithms would need to be made more robust to solve the problem. Finally, the scalability of the system for larger fleets and deliveries in more complex urban environments needs to be examined.

CONCLUSION

This work shows the very large potential that the combination of IoT and CV technologies can have to improve last-mile delivery operations in smart logistics systems. Utilizing real-time data collection, automated visual analysis, and edge computing the presented smart delivery framework resolves fundamental challenges of traditional logistics such as delays, inaccuracies, and environmental inefficiencies. Experimental results showed significant improvements in several performance metrics. Compared with conventional smart delivery systems, the smart delivery system could decrease average delivery time by 15%, increase overall delivery accuracy by 6%, and

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reduce fuel consumption and CO_2 emissions by 12% and 10%, respectively. Such improvements reflect the success of the suggested method in increasing operational performance, promoting environmental sustainability and improving customer satisfaction. In summary, these combined technologies provide a more connected and automated approach to last-mile delivery, enabling smarter, greener, and more efficient services. Moving forward, the introduction of machine learning models for predictive analytics, real-time route optimization, and enhanced anomaly detection may take future work to the next level and would serve to best optimize performance and scalability of the system. The future of last-mile delivery will be filled with exciting developments to be developed in the integrated technology of IoT and Computer Vision to enhance this service. AI and machine learning usage for predictive routing, and the use of autonomous delivery vehicles and drones to minimize human dependency are some of the areas future researchers can focus on. Blockchain technology, increased system scalability, and more eco-friendly supply chain deliveries all contribute to smarter, greener logistics object. Also, enhancing the front end, creating real-time tracking and feedback systems as technology improvements can also improve the quality of service and satisfaction.

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